DPO and GRPO

RL without a critic, and preferences without RL

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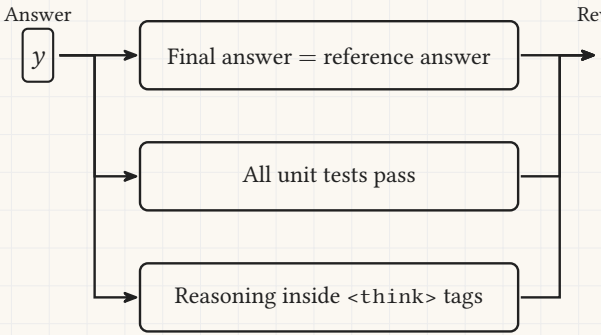
20 October 2026

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Section 1

Verifiable rewards

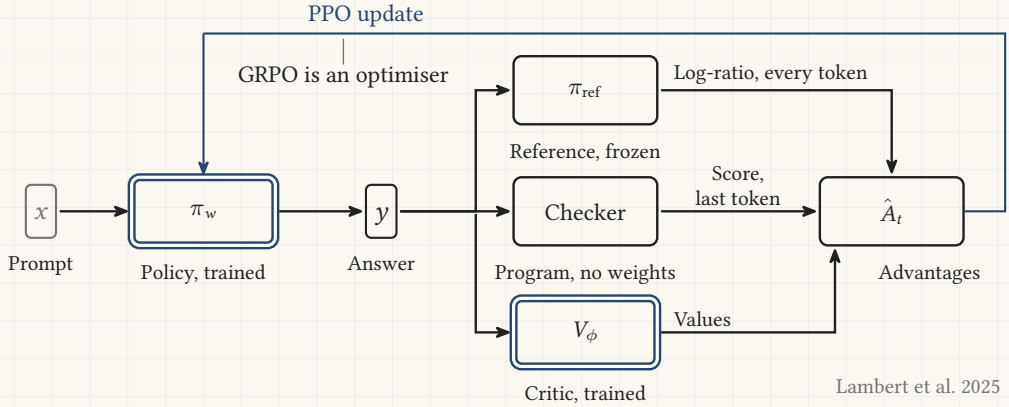
Rewards a program can check



$r \in \{0, 1\}$	Liu et al. 2025
$\in \{-1, 1\}$	Yu et al. 2025
$\in \{0, 10\}$	Lambert et al. 2025
$=$ accuracy $+$ format	Guo et al. 2025

DeepSeek-R1: learned reward models get hacked; retraining costs compute.

The same loop with a checker



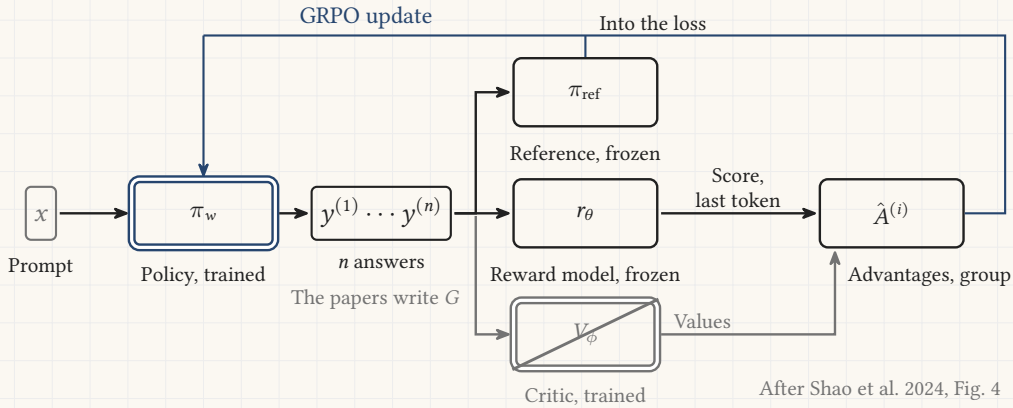
Lambert et al. 2025

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Section 2

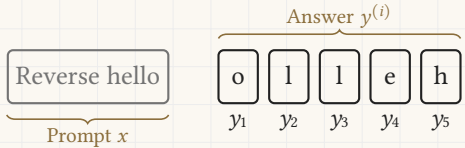
GRPO

From PPO to GRPO



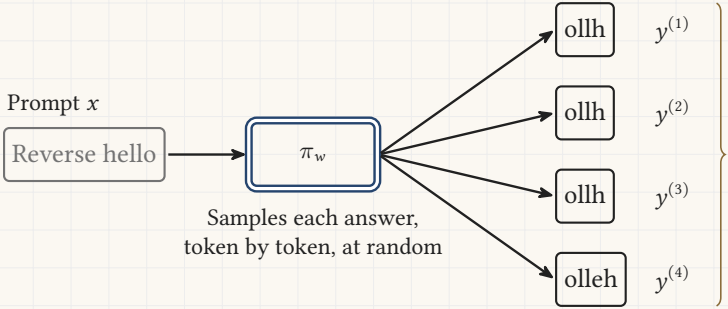
After Shao et al. 2024, Fig. 4

Every token's return is the answer's reward



Reward r_t	0	0	0	0	$r^{(i)}$		Every token of $y^{(i)}$: multiplier
Return $G_t = r_t + \cdots + r_5$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$\gamma = 1$	$G_t - \text{baseline} = r^{(i)} - \text{baseline}$

The group: n answers to the same prompt



GRPO: Group Relative Policy Optimization

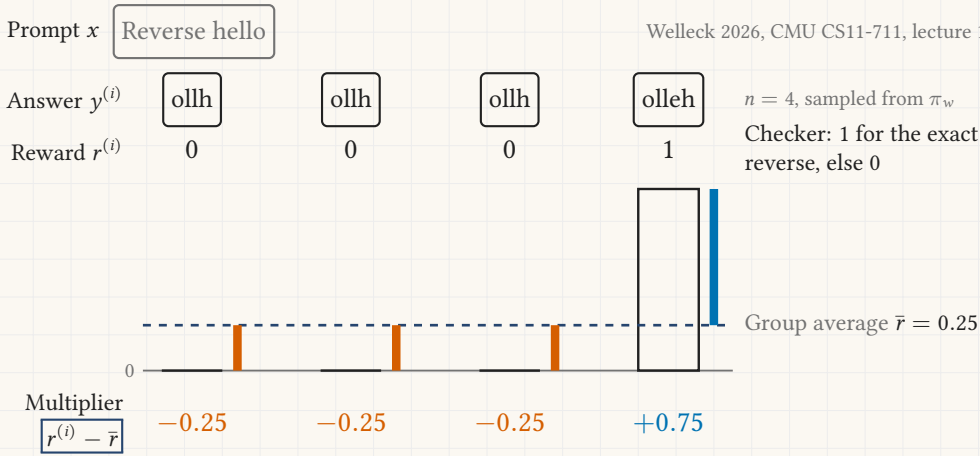
The group of x: its n answers

DeepSeekMath: $n = 64$

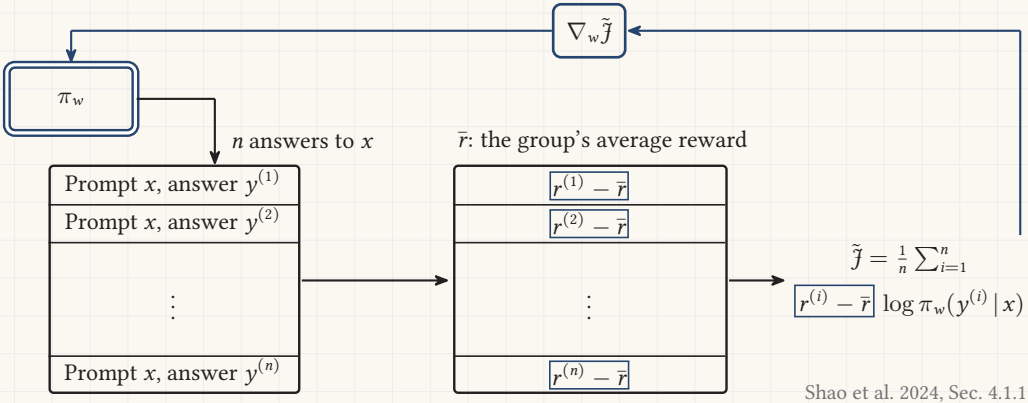
Shao et al. 2024, Sec. 4.2

Welleck 2026, CMU CS11-711, lecture 17

The group average is the baseline



The same loop for GRPO



Shao et al. 2024, Sec. 4.1.1

3

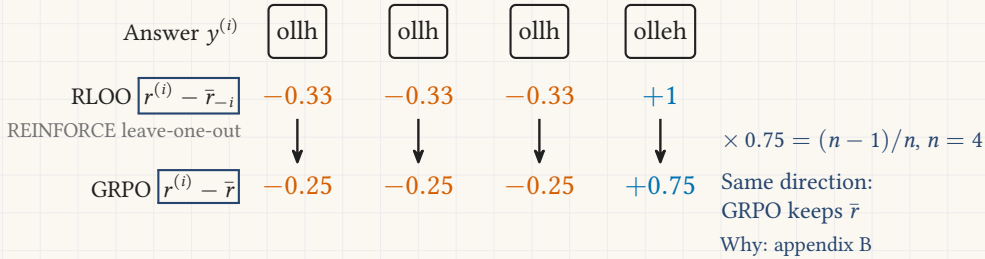
Section 3, optional

GRPO in detail

Leaving the answer out only rescales the multiplier

Morning's rule: no baseline may see the answer it scores

$\bar{r}$ includes it; RLOO uses $\bar{r}_{-i}$, the others' average



Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

The GRPO objective

Shao et al. 2024, eq. (3)

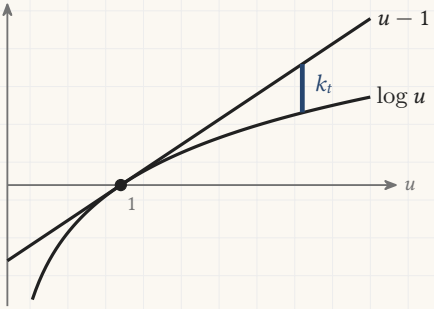
One update per batch:
$\rho_t = 1 \Rightarrow$ policy gradient

Group advantage:
divided by the spread

$$L(w) = \underbrace{\text{average}_i \ \text{average}_t}_{\text{Over answers, then tokens: divided by the answer's length}} \Big[\underbrace{\min\big(\rho_t \, \boxed{\hat{A}^{(i)}}, \ \text{clip}(\rho_t, \, 1-\epsilon, \, 1+\epsilon) \, \boxed{\hat{A}^{(i)}}\big)}_{\text{PPO clipping}} - \underbrace{\beta \, k_t}_{\text{KL estimate, weight } \beta} \Big]$$

The KL estimate in the loss

$$k_t = (u - 1) - \log u \quad u = \pi_{\text{ref}}/\pi_w, \text{ per token}$$



Never negative; unbiased for tokens from π_w

Schulman 2020

Weight β

DeepSeekMath	0.04
DeepSeek-R1-Zero	0.001
	New π_{ref} every 400 steps
Two later variants	0

In the reward: may penalise length

Guo et al. 2025

Two later variants: Dr. GRPO and DAPO

	Dr. GRPO GRPO Done Right	DAPO
Problem	GRPO's two divisions bias it	Agreeing groups teach nothing
Change	Drop both	Drop them, refill the batch
Gain	Same accuracy, shorter wrong answers	AIME 2024: 42 → 50
	Liu et al. 2025	Yu et al. 2025, Table 1

Dividing by the spread removes the reward's units

DeepSeekMath

Shao et al. 2024, Sec. 4.1.2

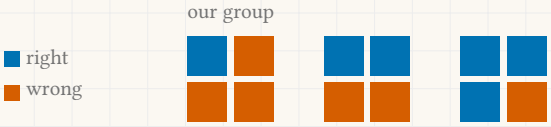
$$\hat{A}^{(i)} = \frac{r^{(i)} - \bar{r}}{\underbrace{\text{std}}_{\text{Spread of the group's rewards}}}$$

Every group gets spread 1:
one step size and one β for any reward

Rewards		Minus $\bar{r}$ (ollh, olleh)	Divided by std: $\hat{A}^{(i)}$	
$r \in \{0, 1\}$	Liu et al. 2025	$-0.25,\ +0.75$	$-0.577,\ +1.732$	
$r \in \{0, 10\}$	Lambert et al. 2025	$-2.50,\ +7.50$	$-0.577,\ +1.732$	
$r \in \{-1, 1\}$	Yu et al. 2025	$-0.50,\ +1.50$	$-0.577,\ +1.732$	std over n

The spread depends only on the fraction right

Three prompts, $n = 4$ answers each, rewards 1 or 0



Right	1 of 4	2 of 4	3 of 4
Fraction right, p	0.25	0.50	0.75
Spread, std	0.433	0.500	0.433

$$\text{std} = \sqrt{p\,(1-p)}$$

rewards 1 or 0, std over n

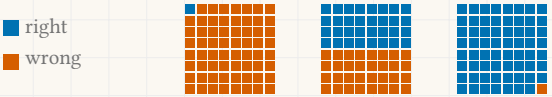
Set by p alone,
and p is how hard the prompt is

few right: hard; most right: easy

Dividing favours prompts whose answers nearly agree

DeepSeekMath's $n = 64$ answers per prompt; a checker's 1 or 0

Shao et al. 2024, Sec. 4.2



Right	1 of 64	32 of 64	63 of 64
Fraction right, p	0.016	0.500	0.984
Spread, std over n	0.124	0.500	0.124
Each multiplier $\times 1/\text{std}$	$\times$ 8.06	$\times$ 2.00	$\times$ 8.06

A difficulty bias

Liu et al. 2025, Sec. 3.1

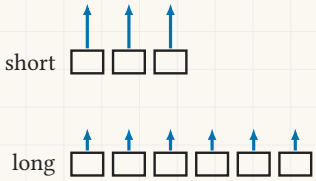
Rewards 1 or 0: no units to remove;
Dr. GRPO drops the division

Dividing by the length favours long wrong answers

GRPO: each token of $y^{(i)}$ gets $\hat{A}^{(i)}/|y^{(i)}|$

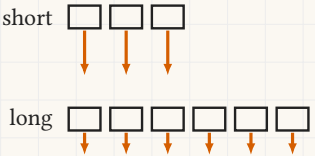
Liu et al. 2025, Sec. 3.1

Right answer, $\hat{A}^{(i)} > 0$



Short ones gain more per token

Wrong answer, $\hat{A}^{(i)} < 0$



Long ones lose less per token:
wrong answers grow longer

Dr. GRPO drops both divisions

Each token of $y^{(i)}$ gets, in GRPO:

$$\left(r^{(i)} - \bar{r}\right) \times \frac{1}{\text{std}} \times \frac{1}{|y^{(i)}|}$$

Difficulty bias Length bias

Liu et al. 2025, Sec. 3.2

In Dr. GRPO:

$$\left(r^{(i)} - \bar{r}\right) \times \frac{1}{C}$$

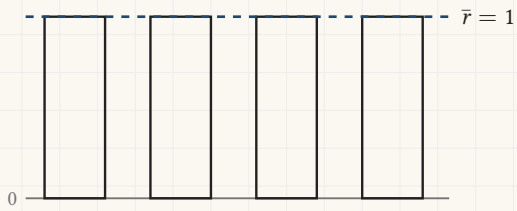
The loop's multiplier, every token alike:
this morning's policy gradient

C: the generation budget, the same for every answer

Groups whose answers all agree teach nothing

Every answer to x correct

Group average $\bar{r} = 1$: every multiplier $r^{(i)} - \bar{r} = 0$



$$\frac{r^{(i)} - \bar{r}}{\text{std}} = \frac{0}{0}$$

Code adds a small δ:

$$\frac{r^{(i)} - \bar{r}}{\text{std} + \delta} = \frac{0}{\delta} = 0$$

The group adds nothing to the step: its n answers were wasted

Five ways to fill the multiplier

Algorithm	Multiplier	Score	In words
REINFORCE	G_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Rest of the episode
With a baseline	$G_t - V(s_t)$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than usual here
Actor-critic	$r_t + \gamma\, V(s_{t+1}) - V(s_t)$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than predicted
PPO	Ψ_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Same, step constrained
GRPO	$(r^{(i)} - \bar{r})/\text{std}$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than its group

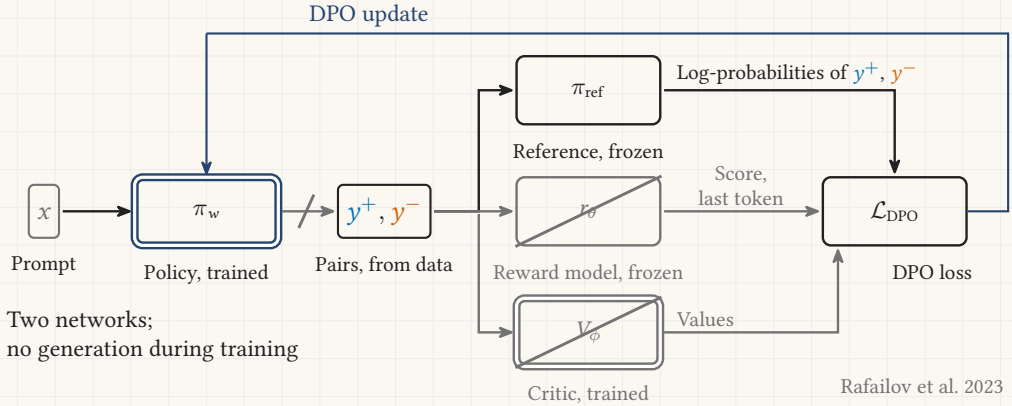
std over n or $n-1$

4

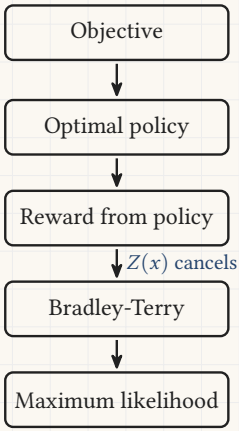
Section 4

Direct preference optimisation

What DPO removes



From Part 1's objective to the DPO loss



$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Answers y drawn from any π

Rafailov et al. 2023, eqs. (3) to (7)

Every step: appendix

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

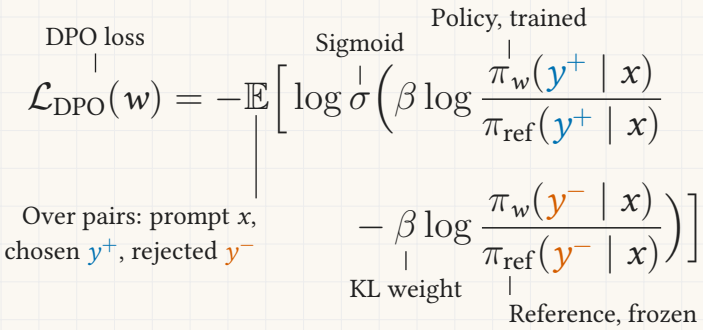
$$P(y^+ \succ y^- \mid x) = \sigma\left(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\mathrm{ref}}(y^+ \mid x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\mathrm{ref}}(y^- \mid x)}\right)$$

$$\mathcal{L}_{\mathrm{DPO}}(w) = -\mathbb{E}\big[\log P_w(y^+ \succ y^- \mid x)\big]$$

P_w: π_w for π^*

The DPO loss

Rafailov et al. 2023, eq. (7)



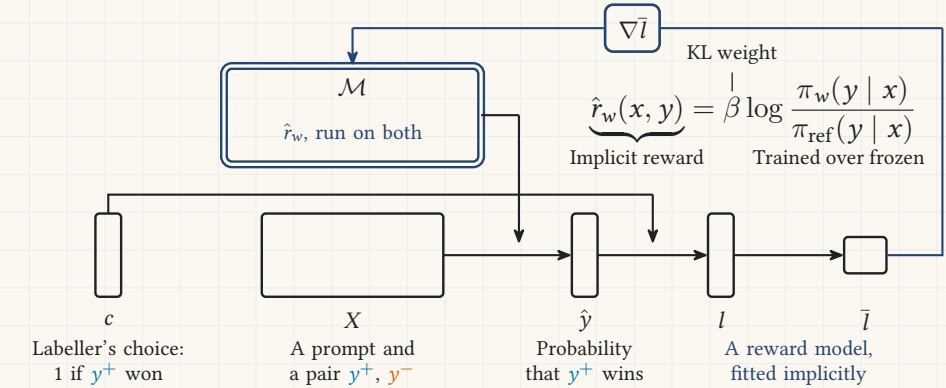
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Section 5

DPO in practice

The DPO loss is a reward-model loss

$$\hat{y} = \sigma\big(\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)\big) \qquad l = -\log \hat{y}$$



References

Wouter Kool, Herke van Hoof and Max Welling. Buy 4 REINFORCE samples, get a baseline for free! *ICLR workshop on deep reinforcement learning meets structured prediction*, 2019. openreview.net/forum?id=r1lgTGL5DE

John Schulman. Approximating KL divergence. Blog post, 2020. joschu.net/blog/kl-approx.html

Rafael Rafailov and others. Direct preference optimization: your language model is secretly a reward model. *NeurIPS*, 2023. arxiv.org/abs/2305.18290

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References, continued

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Daya Guo and others. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature* 645, 633 to 638, 2025. doi.org/10.1038/s41586-025-09422-z

Zichen Liu and others. Understanding R1-Zero-like training: a critical perspective. *COLM*, 2025. arxiv.org/abs/2503.20783

Qiying Yu and others. DAPO: an open-source LLM reinforcement learning system at scale. *NeurIPS*, 2025. arxiv.org/abs/2503.14476

Nathan Lambert and others. Tulu 3: pushing frontiers in open language model post-training. *COLM*, 2025. arxiv.org/abs/2411.15124

Sean Welleck. Reinforcement learning for LLMs. CS11-711 Advanced NLP, lecture 17, Carnegie Mellon University, 17 March 2026. cmu-13.github.io/anlp-spring2026

A

Appendix

Deriving the DPO loss

What the derivation works with

x	One prompt, held fixed
y	An answer: finitely many, bounded length
$\pi(y \mid x)$	Any policy: $\pi(y \mid x) \geq 0$, summing to 1 over y
$\pi_{\text{ref}}(y \mid x)$	The frozen reference: $\pi_{\text{ref}}(y \mid x) > 0$, softmax outputs
$r(x, y)$	Any reward, a number per answer
β	The KL weight: $\beta > 0$
$\mathbb{E}[f(y)]$	$= \sum_y \pi(y \mid x)\, f(y)$: the average over y drawn from π
$\log$	Natural logarithm; $0 \log 0$ counts 0

Step 1: Part 1's objective as sums

Rafailov et al. 2023, eq. (3); App. A.1, eq. (11)

$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\text{ref}}\big)$$

Part 1, one prompt x

$$= \underbrace{\textstyle\sum_y \pi(y \mid x)\, r(x,y)}_{\text{Average reward}} - \beta \underbrace{\textstyle\sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)}}_{\text{KL of } \pi \text{ from } \pi_{\text{ref}}}$$

Averages as sums

Each prompt x has its own $\pi(y \mid x)$: solve one prompt at a time.

Step 2: one sum, one logarithm

Rafailov et al. 2023, App. A.1, eq. (12)

$$\mathcal{J}(\pi) = \sum_y \pi(y \mid x)\left[r(x,y) - \beta \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)}\right]$$ Same weights: one sum

$$= -\beta \sum_y \pi(y \mid x)\left[\log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} - \frac{r(x,y)}{\beta}\right]$$ Take out $-\beta$

$$= -\beta \sum_y \pi(y \mid x)\left[\log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} - \log e^{r(x,y)/\beta}\right]$$ $\frac{r}{\beta} = \log e^{r/\beta}$

$$= -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}$$ $\log a - \log b = \log \frac{a}{b}$

Step 3: reweight the reference, then normalise

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

New policy | Reference | Weight

$$Z(x) = \sum_y \pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}$$

Normaliser

Reference $\pi_{\text{ref}}(y \mid x)$ × Weight $e^{r(x,y)/\beta}$ / $Z(x)$ → New policy $\pi^*(y \mid x)$



$\pi^*(y \mid x) > 0$, as $\pi_{\text{ref}} > 0$

$$\sum_y \pi^*(y \mid x) = \frac{Z(x)}{Z(x)} = 1$$

So π^* is a policy

Step 3: the objective is a KL to π^*

Rafailov et al. 2023, App. A.1, eqs. (12) to (14)

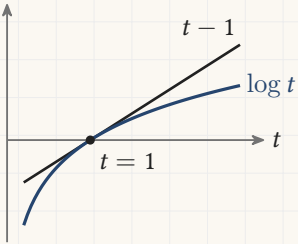
$$\mathcal{J}(\pi) = -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}$$ Step 2

$$= -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{Z(x)\, \pi^*(y \mid x)}$$ Definition of π^*

$$= -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi^*(y \mid x)} + \beta \log Z(x) \sum_y \pi(y \mid x)$$ $\log \frac{a}{bc} = \log \frac{a}{c} - \log b$

$$= -\beta \underbrace{\mathrm{KL}\big(\pi \,\big\|\, \pi^*\big)}_{\text{KL, as in step 1}} + \underbrace{\beta \log Z(x)}_{\text{Same for every } \pi}$$ $\sum_y \pi(y \mid x) = 1$

Gibbs' inequality: a KL is never negative



$t-1-\log t$ has slope $1-1/t$: lowest, 0, at $t=1$.

For policies p, q of one prompt, with every $q(y)>0$:
$\mathrm{KL}(p\,\|\,q)\geq 0$, and $\mathrm{KL}(p\,\|\,q)=0$ only if $p=q$.

$$
\begin{aligned}
-\mathrm{KL}(p\,\|\,q) &= \textstyle\sum_{p(y)>0} p(y)\log\dfrac{q(y)}{p(y)} && \text{Terms with } p(y)=0 \text{ count } 0\\
&\leq \textstyle\sum_{p(y)>0} p(y)\left(\dfrac{q(y)}{p(y)}-1\right) && \log t\leq t-1\\
&= \textstyle\sum_{p(y)>0} q(y)-1 \;\leq\; 0 && \textstyle\sum_{p(y)>0} p(y)=1,\ \sum_{p(y)>0} q(y)\leq 1
\end{aligned}
$$

Equal only if $q(y)=p(y)$ wherever $p(y)>0$, and no $p(y)$ is 0, as every $q(y)>0$: so $q=p$.

Step 3: π^* is the only maximiser

$$\mathcal{J}(\pi) = \beta \log Z(x) - \beta\, \mathrm{KL}\big(\pi \,\big\|\, \pi^*\big) \qquad \text{Step 3, as a KL}$$

$$\leq \beta \log Z(x), \text{ for every policy } \pi \qquad \text{Gibbs } (\pi^* > 0);\ \beta > 0$$

$$\mathcal{J}(\pi) = \beta \log Z(x) \text{ exactly when } \pi = \pi^* \qquad \mathrm{KL}\big(\pi^* \,\big\|\, \pi^*\big) = 0;\ \text{Gibbs: only there}$$

The maximiser: $\pi^*(y \mid x) = \dfrac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$

Rafailov et al. 2023, eq. (4)

Small β sharpens π^*; large β keeps it near π_{ref}.

Step 4: reading the reward off π^*

Rafailov et al. 2023, eq. (5)

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

Step 3

$$\log \pi^*(y \mid x) = \log \pi_{\text{ref}}(y \mid x) + \frac{r(x,y)}{\beta} - \log Z(x)$$

Take logs: every factor > 0

$$r(x,y) = \underbrace{\beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)}}_{\text{Implicit reward}} + \underbrace{\beta \log Z(x)}_{\substack{\text{Same for every} \\ \text{answer to } x}}$$

Solve for r

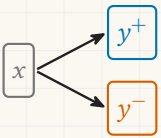
$Z(x)$ sums over all answers: too costly to compute.

Step 5: $Z(x)$ cancels in Bradley-Terry

Rafailov et al. 2023, eqs. (1), (6); App. A.2

$$P(y^+ \succ y^- \mid x) = \sigma\big(r(x, y^+) - r(x, y^-)\big)$$

Assumed: Bradley-Terry, reward r; π^* its maximiser



$$= \sigma\Big(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} + \cancel{\beta \log Z(x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)} - \cancel{\beta \log Z(x)}\Big)$$

Step 4, for y^+ and for y^-

Same prompt x: $Z(x)$ cancels

$$= \sigma\Big(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)}\Big)$$

Step 6: the network's implicit reward

Rafailov et al. 2023, Sec. 4

$$\hat{r}_w(x, y) = \beta \log \frac{\pi_w(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

Define; $\pi_w > 0$, a softmax

$$\pi_{\text{ref}}(y \mid x)\, e^{\hat{r}_w(x,y)/\beta} = \pi_w(y \mid x)$$

$e^{\log a} = a$

$$\pi^*(y \mid x) = \frac{\pi_w(y \mid x)}{\sum_{y'} \pi_w(y' \mid x)} = \pi_w(y \mid x)$$

Optimal for $\hat{r}_w$

Step 3; the sum is 1

$$P_w(y^+ \succ y^- \mid x) = \sigma\big(\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)\big)$$

Predicted

Step 5 with this reward

The network is the optimal policy of its own reward.

Step 6: maximum likelihood on the pairs

Rafailov et al. 2023, eq. (7)

$$P_w(\text{data}) = \prod_{\text{pairs}} P_w(y^+ \succ y^- \mid x)$$

Independent pairs: multiply

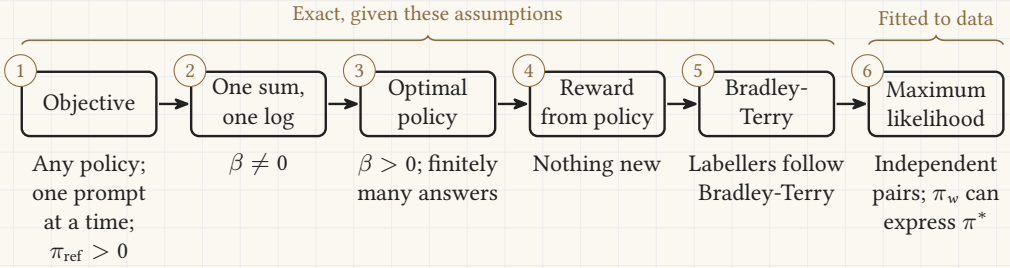
$$\log P_w(\text{data}) = \sum_{\text{pairs}} \log P_w(y^+ \succ y^- \mid x)$$

Log of a product: sum of logs

$$\mathcal{L}_{\text{DPO}}(w) = -\frac{1}{N} \underbrace{\sum_{\text{pairs}}}_{\text{The } N \text{ pairs: the } \mathbb{E} \text{ of the DPO loss}} \log P_w(y^+ \succ y^- \mid x)$$

log increases, and $-\frac{1}{N} < 0$: maximise becomes minimise

What each step assumes



B

Appendix

The group average as a baseline

The group average includes the answer's own reward

This morning's rule

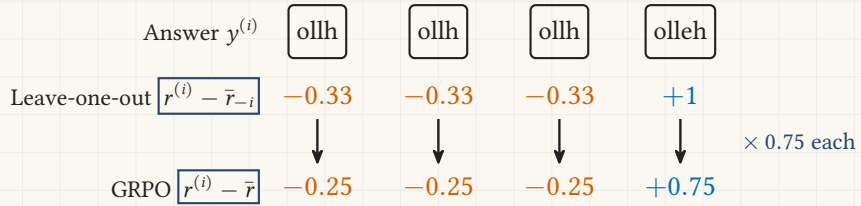
$$\mathbb{E}\big[\, b(x)\, \nabla_w \log \pi_w(y \mid x) \,\big] = b(x)\, \mathbb{E}\big[\nabla_w \log \pi_w(y \mid x)\big] = 0$$

A baseline $b(x)$ may look at the prompt x, never at the answer y

Answer $y^{(i)}$	ollh	ollh	ollh	olleh	
Reward $r^{(i)}$	0	0	0	1	
Group average $\bar{r}$	0.25	0.25	0.25	0.25	Sees $y^{(i)}$: breaks the rule
Others' average $\bar{r}_{-i}$	0.33	0.33	0.33	0	Never sees $y^{(i)}$: keeps it

Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

The group average is the others' average, rescaled



$$r^{(i)} - \bar{r} = \frac{n-1}{n}\left(r^{(i)} - \bar{r}_{-i}\right)$$

since $n\,\bar{r} = r^{(i)} + (n-1)\,\bar{r}_{-i}$

Leave-one-out: on average the policy gradient
GRPO: times $(n-1)/n$; the step size absorbs it

Welleck 2026, CMU CS11-711, lecture 17
Kool et al. 2019; Liu et al. 2025, App. A